

The Intricacies of the System V Semaphore

This article explains the semaphore related data structures and suggests suitable programs to implement the semaphore set.

*S*emaphore is a principal synchronisation mechanism for concurrent access of shared resources. Semaphores are classified as binary and counting semaphores. A binary semaphore is used to implement mutual exclusion and the counting semaphore is used when many shared resources are to be synchronised.

For example, if we want to book a train ticket, the train data is in the central database so that many users can access it at the same time. This is why we have many counters to book our tickets. But if two people are allowed to book a ticket to the same destination, on the same date and for the same train, then it could lead into problems if both receive a ticket with the same seat number. To avoid this, when the first user enters the records to book the ticket, the other is blocked until the first user exits. To accomplish this kind of environment, a binary semaphore is used.

Another approach of semaphores is resource counting. If there are many shared resources to be synchronised, then a counting semaphore is a right choice. Consider an example of a university that announces the results of graduating students on its website. If a count is set to

a specific number, say 1,000, this indicates that only 1,000 students can log into the website concurrently and check their results. Thereafter, any subsequent access will be locked till the count is less than a thousand.

In Linux, the System V Semaphore creates a semaphore set where many semaphores can be created, each being a binary or a counting semaphore, depending on the requirement.

Now let's consider creating a semaphore set with one binary and one counting semaphore. To understand this, let us consider another real world scenario: a team of 10 members working offshore have to take up a conference call from a client who is on-site. They all get connected to the conference call, so we need to have a counting semaphore that will have to maintain a count of the team members and synchronise access to common resources between them. But it has been noticed that during a conference call, only one person is given access to talk at a particular time and the rest are not allowed access. So a binary semaphore is required in order to achieve mutual exclusion amongst the team members.

Considering this scenario, this article explains the semaphore related data structures and suggests suitable programs to implement the semaphore set, which may be of the binary